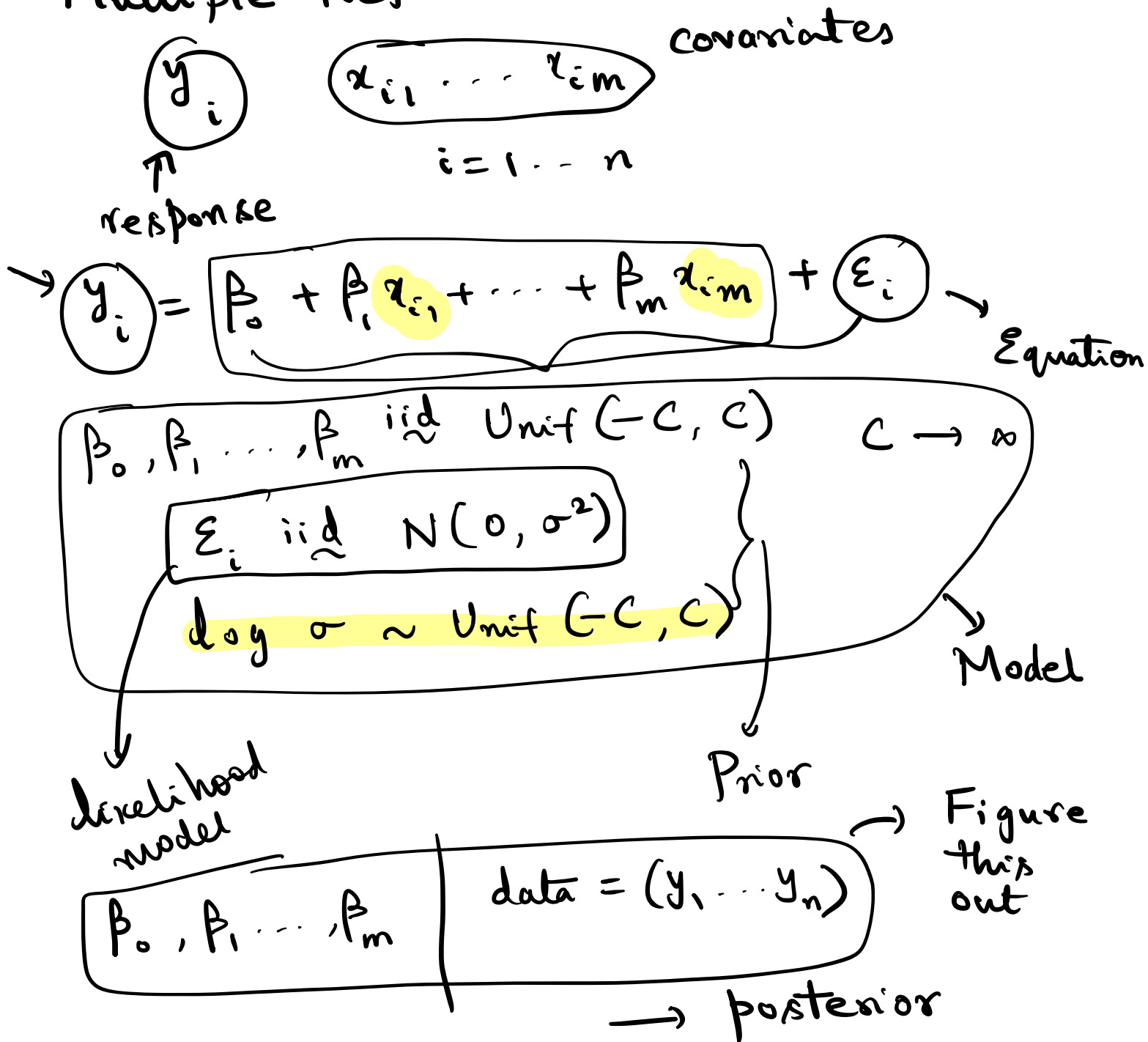


LECTURE FIVE (10 Sept 2026)

153-248

Multiple Regression:



$$f(\beta_0 \dots \beta_m)$$

$$\beta_0, \dots, \beta_m | y_1, \dots, y_n$$

$$= \int_0^\infty f(\beta_0 \dots \beta_m, \sigma) d\sigma$$

$$\beta_0, \dots, \beta_m, \sigma | y_1, \dots, y_n$$

$$\beta, \sigma | y_1, \dots, y_n$$

can be written explicitly
 Integration also explicit. ($C \rightarrow \infty$)

$$f(\beta_0, \dots, \beta_m) \propto \left[\frac{1}{S(\beta)} \right]^{n/2}$$

$$\beta_0, \dots, \beta_m \mid y_1, \dots, y_n$$

$$S(\beta) = S(\beta_0, \beta_1, \dots, \beta_m)$$

$$= \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_m x_{im})^2$$

① Suppose $\hat{\beta}$ minimizes $S(\beta)$
 least squares estimate
 then $f(\beta)$ is maximized at $\hat{\beta}$.
 $\beta \mid \text{data}$

$$\textcircled{2} f(\beta) \propto \left(\frac{1}{S(\beta)} \right)^{n/2} \rightarrow$$

$$\propto \left(\frac{S(\hat{\beta})}{S(\beta)} \right)^{n/2} \rightarrow$$

$$f(\beta) \propto \left(\frac{S(\hat{\beta})}{S(\beta)} \right)^{n/2}$$

$(\beta_0, \dots, \beta_m)$

maximized at $\beta = \hat{\beta}$

when n is large, this is highly

concentrated around the least squares estimator.

Connection to the multivariate t-density

t-density: $x = (x_1, \dots, x_p)$, ν : degrees of freedom

$$\propto \left[\frac{1}{1 + \frac{1}{\nu} (x - \mu)^T \Sigma^{-1} (x - \mu)} \right]^{\frac{\nu + p}{2}} \left(\frac{S(\hat{\beta})}{S(\beta)} \right)^{n/2}$$

$t_p(\mu, \Sigma, \nu)$

$\leftarrow$
 $\left(\frac{S(\hat{\beta})}{S(\beta)} \right)^{n/2}$

Normal Density

$$\propto \exp \left[- \frac{(x - \mu)^T \Sigma^{-1} (x - \mu)}{2} \right]$$

$\exp \left[- \underbrace{\text{square function}} \right]$

Notation: $N_p(\mu, \Sigma)$

(a) $X \sim N_p(\mu, \Sigma)$ then $\mu = \begin{pmatrix} \mathbb{E} X_1 \\ \mathbb{E} X_2 \\ \vdots \\ \mathbb{E} X_p \end{pmatrix}$

$$\Sigma = \begin{bmatrix} \text{var}(x_1) & & \\ & \text{var}(x_2) & \\ & \text{Cov}(x_i, x_j) & \ddots \\ & & & \text{var}(x_p) \end{bmatrix}$$

$$\Sigma(i, j) = \text{Cov}(X_i, X_j)$$

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_5 \end{pmatrix} \sim N_5(\mu, \Sigma)$$

$$\rightarrow \text{var}(x_4) = \Sigma(4, 4)$$

⑥ $X \sim N_p(\mu, \Sigma) \rightarrow$ all components of X are normal

$$\Rightarrow X_2 \sim N(\mu_2, \Sigma(2,2))$$

$$X_j \sim N(\mu_j, \Sigma(i,j))$$

$\rightarrow$ every linear combination of $X_1, \dots, X_p$ is normal

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N_2\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma(1,1) & \Sigma(1,2) \\ \Sigma(2,1) & \Sigma(2,2) \end{pmatrix}\right)$$

t is Normal

Suppose $X \sim N_p(\mu, \Sigma)$

$$X = \mu + \underbrace{(X - \mu)}$$

Define

$$T = \mu + \frac{X - \mu}{\sqrt{V/v}} \quad \text{where } V \sim \chi^2_v$$

chi-squared χ^2_v

$$Z_1^2 + \dots + Z_v^2$$

$$Z_j \sim N(0,1)$$

$$\mathbb{E}(\chi^2_v) = v$$

$$T = \mu + \frac{X - \mu}{\sqrt{\frac{Z_1^2 + \dots + Z_v^2}{v}}}$$

$$\mathbb{E}(Z_j^2) = 1$$

Fact: $T \sim t_p(\mu, \Sigma, v)$

$T_j \rightarrow j^{\text{th}}$ component of T

$$T = \mu + \frac{X - \mu}{\sqrt{\frac{z_1^2 + \dots + z_v^2}{v}}}$$

$$T_j = \mu_j + \frac{X_j - \mu_j}{\sqrt{\frac{z_1^2 + \dots + z_v^2}{v}}} \sim t_1(\mu_j, \Sigma(\omega, \lambda), v)$$

Individual components of Multivariate t are univariate t .

Back to Linear Regression

$$f(\beta | \text{data}) \propto \left(\frac{S(\hat{\beta})}{S(\beta)} \right)^{n/2} \quad \beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_m \end{pmatrix}$$

$$S(\beta) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_m x_{im})^2$$

$$\left[\frac{1}{1 + \frac{1}{v} (x - \mu)^T \Sigma^{-1} (x - \mu)} \right]^{\frac{v+p}{2}} \quad v, \mu, \Sigma$$

$(p = m+1)$

$$n = v + p \Rightarrow v = n - p = n - m - 1$$

$$S(\beta) = \|y - X\beta\|^2$$

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1m} \\ \vdots & x_{21} & x_{22} & & x_{2m} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & & x_{nm} \end{bmatrix}$$

$$\beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{pmatrix}$$

$$S(\beta) = \|y - X\beta\|^2$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

→ least squares

→ minimizes $S(\beta)$

$$\stackrel{\text{Pythagorean Identity}}{=} \|y - X\hat{\beta} + X\hat{\beta} - X\beta\|^2 = \|y - X\hat{\beta}\|^2 + \|X\hat{\beta} - X\beta\|^2$$

& the cross product is zero

$$(a+b)^2 = a^2 + b^2 + 2ab$$

$$S(\beta) = \|y - X\hat{\beta}\|^2 + \|X\hat{\beta} - X\beta\|^2$$

$$X^T X \hat{\beta} = X^T y$$

$$\|x\|^2 = x^T x$$

$$= S(\hat{\beta}) + (\hat{\beta} - \beta)^T X^T X (\hat{\beta} - \beta)$$

$$= S(\hat{\beta}) + (\beta - \hat{\beta})^T X^T X (\beta - \hat{\beta})$$

$$f_{\beta|\text{data}}(\beta) \propto \left[\frac{S(\hat{\beta})}{S(\beta)} \right]^{n/2}$$

$$= \left[\frac{S(\hat{\beta})}{S(\hat{\beta}) + (\beta - \hat{\beta})^T X^T X (\beta - \hat{\beta})} \right]^{n/2}$$

$$p = m + 1$$

$$v = n - m - 1$$

$$\left[\frac{1}{1 + \frac{1}{v} (\alpha - \mu)^T \Sigma^{-1} (\alpha - \mu)} \right]^{\frac{v+p}{2}}$$

$$\left(1 + \frac{1}{S(\hat{\beta})} (\beta - \hat{\beta})^T X^T X (\beta - \hat{\beta}) \right)^{n/2}$$

$$x \leftrightarrow \beta, \mu = \hat{\beta} \quad \frac{\Sigma^{-1}}{\nu} = \frac{X^T X}{S(\hat{\beta})}$$

$$\Rightarrow \Sigma^{-1} = \nu \frac{X^T X}{S(\hat{\beta})} = (n-m-1) \frac{X^T X}{S(\hat{\beta})}$$

$$\Rightarrow \Sigma = \frac{S(\hat{\beta})}{n-m-1} (X^T X)^{-1}$$

We derived

$$\beta | \text{data} \sim t_{m+1} \left(\hat{\beta}, \frac{S(\hat{\beta})}{n-m-1} (X^T X)^{-1}, n-m-1 \right)$$

$$S(\hat{\beta}) = \text{smallest value of } S(\beta)$$

$$= \sum_{i=1}^n \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_m x_{im} \right)^2$$

$$= \text{RSS}$$

RESIDUAL SUM OF SQUARES

$$\frac{S(\hat{\beta})}{n-m-1} = \frac{\text{RSS}}{n-m-1} \doteq \hat{\sigma}_{\text{unbiased}}^2$$

$$\beta | \text{data} \sim t_{m+1} \left(\hat{\beta}, \hat{\sigma}_{\text{unbiased}}^2 (X^T X)^{-1}, n-m-1 \right)$$

$$\beta_j | \text{data} \sim t(\hat{\beta}_j, \hat{\sigma}_2^2 \text{unbiased} (X^T X)^{-1}(j+1, j+1), n-m-1)$$

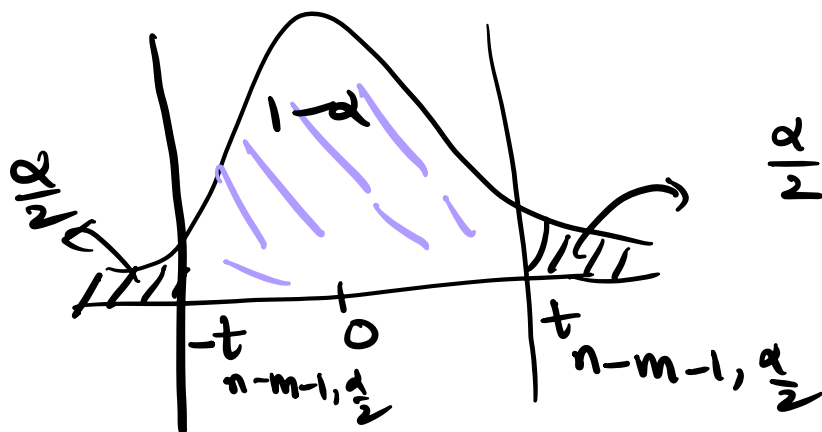
standard error of β_j : $\sqrt{\hat{\sigma}_2^2 \text{unbiased} (X^T X)^{-1}(j+1, j+1)}$



Interval:

$$P\left\{ \beta_j \in \left[\hat{\beta}_j - s.e(\hat{\beta}_j) t_{n-m-1, \frac{\alpha}{2}}, \hat{\beta}_j + s.e(\hat{\beta}_j) t_{n-m-1, \frac{\alpha}{2}} \right] \mid \text{data} \right\} = 1 - \alpha$$

$$\frac{\beta_j - \hat{\beta}_j}{s.e(\hat{\beta}_j)} \mid \text{data} \sim t_{n-m-1}$$



$$P \left[-t_{n-m-1, \frac{\alpha}{2}} \leq \frac{\beta_j - \hat{\beta}_j}{s.e(\hat{\beta}_j)} \leq t_{n-m-1, \frac{\alpha}{2}} \mid \text{data} \right] = 1 - \alpha$$

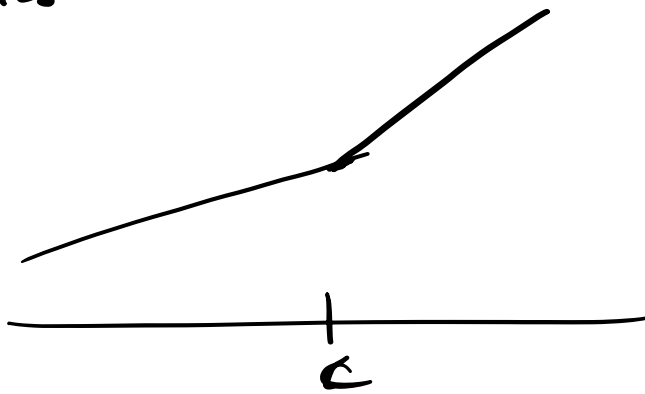
① Linear Regression

② Non Linear Regression

$$y_t = \beta_0 + \beta_1 t + \beta_2 \text{ReLU}(t-c) + \varepsilon_t$$

observed time series

$$\text{ReLU}(t-c) = \max(t-c, 0) = (t-c)_+$$



If c is fixed, then this is linear regression

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \quad X = \begin{bmatrix} 1 & 1 & \text{ReLU}(1-c) \\ \vdots & \vdots & \vdots \\ 1 & n & \text{ReLU}(n-c) \end{bmatrix}$$

c unknown $\rightarrow$ nonlinear regression.

② Sinusoid

$$y_t = \beta_0 + \beta_1 \cos(\omega t) + \beta_2 \sin(\omega t) + \varepsilon_t$$

ω known $\rightarrow$ multiple linear regression

$$X = \begin{bmatrix} 1 & \cos(\omega_1) & \sin(\omega_1) \\ \vdots & \cos(\omega_2) & \sin(\omega_2) \\ \vdots & \vdots & \vdots \\ 1 & \cos(\omega_n) & \sin(\omega_n) \end{bmatrix}$$

$$y_t = \beta_0 + \beta_1 \cos(\omega_1 t) + \beta_2 \sin(\omega_1 t) + \beta_3 \cos(\omega_2 t) + \beta_4 \sin(\omega_2 t) + \epsilon_t$$

DFT

FFT

Periodogram

Spectral Analysis

Signal Processing